

# Strings

15 Interview Questions — Easy to Hard

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**The Complete DSA + SQL Interview Preparation Series**

Book 2 of 13

For placement preparation, coding interviews, and product-based company rounds.

## About This Book

This book covers string manipulation, the second-most common category of coding interview questions after arrays. String problems test your comfort with character frequency, two pointers, sliding windows over text, and classic pattern-matching algorithms.

Unlike the first book in this series, questions here are kept short and direct: just the problem, and a sample input/output pair to show you exactly what's expected. No hints, no constraints tables, no scaffolding — the idea is to read the problem the way it would sound in a live interview and start reasoning immediately.

## How To Use This Book

Read each problem once. Restate it in your own words before writing any code. Use the sample input/output only to confirm your understanding of the expected behavior — not as a spec to reverse-engineer from. Try to solve every problem in your head or on paper before typing anything.

## Difficulty Breakdown

| Difficulty | Count |
|------------|-------|
| Easy       | 2     |
| Medium     | 8     |
| Hard       | 5     |

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## Q1. The Mirror Word

Easy

Topic: Palindrome

A support ticket system flags certain usernames as 'mirror IDs' if they read exactly the same forwards and backwards. Write a check that tells whether a given username is a mirror ID.

INPUT

madam

OUTPUT

true

## Q2. Flip the Sentence

Easy

Topic: Reverse Words

A messaging app has a 'reverse mode' prank filter that reverses the order of words in a message (but not the letters within each word), collapsing any extra spaces along the way.

INPUT

the sky is clear

OUTPUT

clear is sky the

## Q3. Same Letters, Different Order

Medium

Topic: Anagrams

A word-game app needs to check whether two submitted words are anagrams of each other — made of exactly the same letters, just rearranged — to award a bonus round.

INPUT

listen, silent

OUTPUT

true

## Q4. The Loudest Character

Medium

Topic: Character Frequency

A text-analytics tool scans a block of customer feedback and needs to report the character that appears most frequently, ignoring spaces. If there's a tie, return the one that appears first in the text.

INPUT

great support experience

OUTPUT

e

## Q5. Compress the Log Line

Medium

Topic: String Compression

A logging system compresses repeated consecutive characters in a status string to save disk space, replacing runs with the character followed by its count. If the compressed version isn't shorter, keep the original.

INPUT

aaabbcccd

OUTPUT

a3b2c4d1

## Q6. The Longest Unique Stretch

Medium

Topic: Longest Substring

A password-strength checker looks for the longest stretch of a candidate password that has no repeating characters, since long unique runs are considered a strength signal.

INPUT

pwwkew

OUTPUT

3 (the substring "wke")

## Q7. Smallest Window with All Ingredients

Hard

Topic: Minimum Window Substring

A recipe-search engine, given a long block of recipe text and a set of required ingredient letters, must find the smallest contiguous window of the text that contains every required character (including duplicates), or report that no such window exists.

INPUT

```
text = "ADOBECODEBANC", required =  
"ABC"
```

OUTPUT

"BANC"

## Q8. Unpack the Archive Code

Hard

Topic: Decode String

A file-archiving format encodes repeated blocks as `k[encoded_string]`, meaning the `encoded_string` inside the brackets is repeated `k` times, and encodings can be nested. Decode a given archive code back into the original string.

INPUT

`3[a2[c]]`

OUTPUT

`accaccacc`

## Q9. Strip the Noise Characters

Easy

Topic: Remove Duplicates

A data-cleaning step removes consecutive duplicate characters from a scraped string, collapsing every run of the same character down to a single occurrence.

INPUT

`aaabbbccd`

OUTPUT

`abcd`

## Q10. Is the Bracket Sequence Safe to Deploy?

Medium

Topic: Balanced Parentheses

A config-file linter checks whether every opening bracket in a line — `()`, `{}`, `[]` — has a matching, correctly nested closing bracket before the line is accepted into the build.

INPUT

`{[()]}`

OUTPUT

`true`

## Q11. Find the Leaked Snippet

Hard

Topic: Pattern Matching

A plagiarism-detection tool needs to find every starting index in a large document where a given short pattern string appears exactly, without relying on a built-in substring search function.

### INPUT

```
text = "ababcababcabc", pattern =  
"abc"
```

### OUTPUT

```
[2, 7, 10]
```

## Q12. The Repeating Fingerprint

Hard

Topic: Rabin-Karp

A document-similarity service needs to quickly check whether a known 'fingerprint' string occurs anywhere inside a much larger document, using a rolling hash so it doesn't have to re-compare characters at every position from scratch.

### INPUT

```
document = "abxabcabcaby",  
fingerprint = "abcaby"
```

### OUTPUT

```
true, starting at index 6
```

## Q13. Cheapest Way to Fix a Typo Chain

Hard

Topic: Edit Distance

An autocorrect engine needs the minimum number of single-character insertions, deletions, or substitutions required to turn a misspelled word into the closest dictionary word.

### INPUT

```
word1 = "horse", word2 = "ros"
```

### OUTPUT

```
3
```

## Q14. Is This a Rotated Access Key?

Medium

Topic: String Rotation

A security audit needs to check whether one access key string is simply a rotation of another access key string (i.e., some prefix moved to the end), which would indicate they were generated from the same source.

### INPUT

```
key1 = "waterbottle", key2 =  
"erbottlewat"
```

### OUTPUT

```
true
```

## Q15. Longest Shared Boarding Code

Medium

Topic: Pattern Matching

Two airline systems generate boarding reference codes independently. Given two strings, find the length of the longest substring that appears in both, which the fraud team uses to flag suspiciously similar codes.

### INPUT

```
s1 = "flightaxbc", s2 =  
"trainaxbcd"
```

### OUTPUT

```
4 ("axbc")
```